

Negative Exponent Law

MA.8.NSO.1.3

I can rewrite a power with a negative exponent using a positive exponent, and evaluate it.

Yesterday the exponents stopped at 0.

Today they keep going — and 5^{-2} is not a negative number.

Warm-Up

On your own. Four minutes. No notes.

1. Find the value: $\frac{5^4 \cdot 5^0}{5^2}$
2. Which is greater, $\sqrt{30}$ or 5.5?
3. Solve $-3x = 21$
4. What is the reciprocal of $\frac{3}{4}$? What is the opposite of $\frac{3}{4}$?

Warm-Up

Answers.

1. Find the value: $\frac{5^4 \cdot 5^0}{5^2}$ **25**
2. Which is greater, $\sqrt{30}$ or 5.5? **5.5**
3. Solve $-3x = 21$ **$x = -7$**
4. What is the reciprocal of $\frac{3}{4}$? What is the opposite of $\frac{3}{4}$?
 $\frac{4}{3}$ and $-\frac{3}{4}$

Notes

Copy the table.

I. The pattern keeps going past zero

2^4	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}	2^{-3}
16	8	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$

$$2^{-1} = \frac{1}{2^1}$$

$$2^{-2} = \frac{1}{2^2} = \frac{1}{4}$$

$$2^{-3} = \frac{1}{2^3} = \frac{1}{8}$$

Notes

II. The negative exponent law

$$a^{-n} = \frac{1}{a^n} \quad \frac{1}{a^{-n}} = a^n \quad (a \neq 0)$$

$$5^{-2} = \frac{1}{5^2} = \frac{1}{25} \quad \text{but} \quad -5^2 = -25 \quad \text{and} \quad (-5)^2 = 25$$

A negative exponent means reciprocal. It never means a negative value.

The base keeps its own sign: $(-2)^{-3} = \frac{1}{(-2)^3} = \frac{1}{-8} = -\frac{1}{8}$

Notes

III. The laws still work with negative exponents

$$6^{-2} \cdot 6^5 = \frac{1}{6^2} \cdot 6^5 = \frac{7776}{36} = 216$$

$$\text{or } 6^{-2} \cdot 6^5 = 6^{-2+5} = 6^3 = 216$$

$$\frac{9^{-5}}{9^3} = 9^{-5-3} = 9^{-8} = \frac{1}{9^8}$$

Add, subtract and multiply exponents exactly as before — the exponents are just integers now.

Example 1

Negative exponents top and bottom.

$$\frac{2^{-3}}{9^{-2}}$$

Rewrite with positive exponents, then find the value.

Example 1

Worked.

$$\frac{2^{-3}}{9^{-2}} = \frac{9^2}{2^3} \quad \text{each power swaps sides and loses the minus}$$

$$\frac{9^2}{2^3} = \frac{81}{8} = 10.125 \quad \text{evaluate}$$

Answer: $81/8 = 10.125$

Your Turn

Same steps, your numbers. Boards up when done.

$$\frac{3^{-2}}{5^{-1}}$$

Your Turn

Answer.

$$\frac{3^{-2}}{5^{-1}}$$

swap sides, positive exponents: 5^1 over 3^2

Answer: $\frac{5}{9}$

Example 2

A quotient law crossing zero.

$$\frac{7^0}{7^{-4}}$$

Find the value.

Example 2

Worked both ways.

$$\frac{7^0}{7^{-4}} = 7^{0 - (-4)} = 7^4 \quad \text{quotient of powers: } 0 - (-4) = 4$$

$$\frac{7^0}{7^{-4}} = \frac{1}{7^{-4}} = 7^4 \quad \text{zero exponent, then move } 7^{-4} \text{ up}$$

$$7^4 = 2401 \quad \text{evaluate}$$

Answer: 2,401

Your Turn

Same steps, your numbers. Boards up when done.

$$\frac{(-8)^{-4}}{(-8)^{-6}}$$

Your Turn

Answer.

$$\frac{(-8)^{-4}}{(-8)^{-6}}$$

$$-4 - (-6) = 2, \text{ so } (-8)^2$$

Answer: 64

Whiteboards • Question 1 of 9

Boards up on three.

$$8^{-3}$$

Answer it.

Rewrite with a positive exponent.

Whiteboards • Question 1 of 9

Answer.

$$8^{-3}$$

reciprocal of 8^3

Answer: $\frac{1}{8^3}$

Whiteboards • Question 2 of 9

Boards up on three.

A sheet of paper 30 centimeters long is folded in half 7 times.

Each fold makes the top layer half as big.

What fraction of the whole sheet is the top layer after 7 folds?

Answer it.

Whiteboards • Question 2 of 9

Answer.

A sheet of paper 30 centimeters long is folded in half 7 times.

Each fold makes the top layer half as big.

What fraction of the whole sheet is the top layer after 7 folds?

halved 7 times: $2^{-7} = 1 \text{ over } 2^7$

Answer: $\frac{1}{128}$

Whiteboards • Question 3 of 9

Boards up on three.

$$\frac{1}{10^{-2}}$$

Answer it.

Rewrite with a positive exponent, then find the value.

Whiteboards • Question 3 of 9

Answer.

$$\frac{1}{10^{-2}}$$

the power moves up: 10^2

Answer: $10^2 = 100$

Whiteboards • Question 4 of 9

Boards up on three.

$$(-2)^{-3}$$

Answer it.

Find the value.

Whiteboards • Question 4 of 9

Answer.

$$(-2)^{-3}$$

$$1 \text{ over } (-2)^3 = 1 \text{ over } -8$$

$$\text{Answer: } -\frac{1}{8}$$

Whiteboards • Question 5 of 9

Boards up on three.

$$5^{-2}$$

Which is the value?

A. 1/10

B. 1/25

C. -25

D. -10

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$5^{-2}$$

Which is the value?

A. 1/10

B. 1/25

C. -25

D. -10

Answer: B — 1/25

Whiteboards • Question 6 of 9

Boards up on three.

$$6^{-2} \cdot 6^5$$

Which expression is equivalent?

A. 6^{-10}

B. 36^3

C. 6^{-7}

D. 6^3

Answer it.

Whiteboards · Question 6 of 9

Answer.

$$6^{-2} \cdot 6^5$$

Which expression is equivalent?

A. 6^{-10}

B. 36^3

C. 6^{-7}

D. 6^3

Answer: D — 6^3

Whiteboards • Question 7 of 9

Boards up on three.

$$3^{-4} \cdot 3^4$$

Answer it.

Find the value.

Whiteboards · Question 7 of 9

Answer.

$$3^{-4} \cdot 3^4$$
$$3^0$$

Answer: 1

Whiteboards • Question 8 of 9

Boards up on three.

$$\frac{9^{-5}}{9^3}$$

Answer it.

Write it as one power with a positive exponent.

Whiteboards • Question 8 of 9

Answer.

$$\frac{9^{-5}}{9^3}$$
$$9^{-8}$$

Answer: $\frac{1}{9^8}$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Ellen says the value of 5^{-2} is -25 .

Explain why she is wrong, and show the correct value.

Write your answer in sentences.

Use the word reciprocal.

Whiteboards • Question 9 of 9

Answer.

Ellen says the value of 5^{-2} is -25 .

Explain why she is wrong, and show the correct value.

reciprocal, not opposite

Answer: $5^{-2} = 1/5^2 = 1/25$

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Understanding negative exponents — YBB

Evaluate powers with negative exponents — WGS